Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: *Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT* **Paper -1(Adv-2019-P1-Model) Date: 05-04-2026**

Time: 09.00Am to 12.00Pm **GTA-22** **Max. Marks: 186**

KEY SHEET

PHYSICS

1	**B**	2	**C**	3	**D**	4	**A**	5	**BCD**	6	**AC**
7	**AC**	8	**AB**	9	**C**	10	**B**	11	**CD**	12	**AC**
13	**80**	14	**8**	15	**2**	16	**4**	17	**3**	18	**152**

CHEMISTRY

19	**B**	20	**B**	21	**C**	22	**D**	23	**BCD**	24	**ABC**
25	**ABCD**	26	**ABC**	27	**BD**	28	**ABCD**	29	**ACD**	30	**ABC**
31	**5**	32	**43**	33	**3**	34	**8**	35	**8**	36	**8**

MATHEMATICS

37	**D**	38	**C**	39	**C**	40	**C**	41	**ABC**	42	**AC**
43	**AD**	44	**AD**	45	**AC**	46	**AD**	47	**ABCD**	48	**AD**
49	**999**	50	**2**	51	**3**	52	**8**	53	**2**	54	**0**

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SOLUTIONS
PHYSICS

1. Cut off wavelength $\lambda_c = \frac{hc}{eV}$ where V is the accelerating voltage.

   Characteristic wave length $\lambda \propto \frac{1}{(Z-b)^2}$

2. Apply Biot Savart's law $B = B_1 + B_2$

   B_1 (Due to semi-infinite straight wire) $= \frac{\mu_0 l}{4\pi a}$

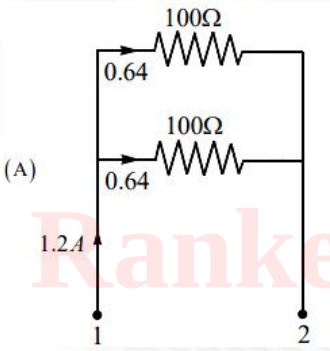
   B_2 (Due to parabolic portion) $= \frac{\mu_0 l}{8a} B = \frac{\mu_0 l}{4a}\left[\frac{1}{\pi} + \frac{1}{2}\right]$

3. $mg - T = ma$

   $T - F = ma \quad F = Bil$

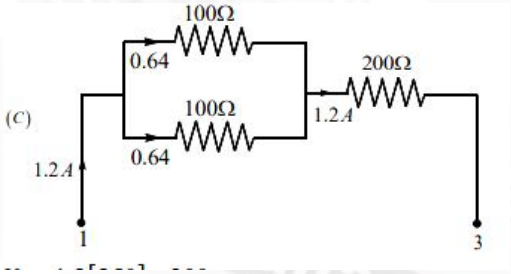
4. $dR = \frac{\rho dx}{A}$

5. Internal force can also change the kinetic energy of system.

6. 

   $30 units \rightarrow 1.2A$

   $1 unit \rightarrow 0.04A$

   $V_{13} = 1.2[250] = 300$

   $30 units \rightarrow 300 Volt$

   $1 unit \rightarrow 10 volt$

7. Flux through a Gaussian surface (cylindrical shape) of length l and radius 2R

   $\oint \vec{E}.d\vec{s} + 2\oint \vec{E}.d\vec{s} = \frac{\lambda l + (-\sigma) 2\pi R l}{\varepsilon_0}$

   Total flux $= \frac{l}{\varepsilon_0}[\lambda - \sigma 2\pi R] = 0 \qquad \sigma = \frac{\lambda}{2\pi R}$

   $i_{RS} = \frac{i}{4} = \frac{mgS}{15eP}$

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8. $|F| = \frac{dU}{dr} = \frac{ke^2}{r^4}$ $\frac{ke^2}{r^4} = \frac{mv2}{r}$ ..... (1)

And $mvr = \frac{nh}{2\pi}$ (2)

By II and I $r = \frac{ke^2 4\pi m}{n^2 h^2} = k_1 \frac{m}{n^2}$

Total energy $= -\frac{1}{2}$ potential energy

$= \frac{-ke^2}{6r^3} = \frac{-ke^2 n^6}{6k_1^3 m^3}$

$\therefore$ Total energy αn^6

Total energy αm^{-3}

9. Use steady state condition

10. $\delta_m = 2i - A = A$ $\Rightarrow i_1 = i_2 = A$ and $r_1 = r_2 = \frac{A}{2}$ $\Rightarrow r_1 = \frac{i_1}{2}$

Using smell's law $\sin A = \mu \sin \frac{A}{2}$ $\Rightarrow A = 2\cos^{-1}\left(\frac{\mu}{2}\right)$

At the second surface $\Rightarrow i_2 = \frac{\pi}{2} \Rightarrow r_2 = C$

$\therefore r_1 = A - C \Rightarrow \sin i_1 = \mu \sin(A - C)$

11. $dI = jdA$ $\Rightarrow dI = kr^2(2\pi r dr)$

$\Rightarrow I = 2\pi K \int_0^a r^3 dr \Rightarrow I = \frac{\pi k a^4}{2}$

Further, field for r>a is $B = \frac{\mu_0 i}{2\pi r} = \frac{\mu_0 k a^4}{4r}$

And field for r<a is $B = \frac{\mu_0\left(\frac{\pi k r^4}{2}\right)}{2\pi r} = \frac{1}{4}\mu_0 k r^3$

12. In case of pure rolling

$f = \frac{mg\sin\theta}{1 + \frac{mR^2}{I}}$ (upward) $\therefore f \alpha \sin\theta$

13. $\frac{l}{2\upsilon} + \frac{\sqrt{4r^2 - l^2}}{\upsilon} = 80$

14. $F = 2mnA\left[(v_0 + v)^2 - (v_0 - v)^2\right]$

$= 2\times10^{-20}\times10^{25}\times1\left[(5+2)^2 - (5-2)^2\right] = 8\text{N}$

15. Since the boards are perpendicular so the impulses provided by board does not effect the motions Perpendicular to the board. Let T_A be the time internal between successive collisions with board A and from board B is

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$$T_B.T_A = \sqrt{\frac{8a}{g\cos a}}; T_B = \sqrt{\frac{8b}{g\cos a}}$$

Let the ball collides X ties with wall B form each time it bounces with wall A

$$xT_B = T_A = 1x = \sqrt{\frac{a}{b}}\tan a = \sqrt{\frac{4}{1}}\times 1 = 2$$

16. $P_1 = 1.0atm, V_1 = 1.0L, T_1 = 122K$

$P_2 = 1.0atm, V_2 = 2.0L, T_2 = 244K, K = 2T_1$

$P_3 = 2.0atm, V_3 = 2.0L, T_3 = 488K, K = 4T_1$

$P_4 = 0.0157atm, V_4 = 64L, T_4 = 122K, K = T_1$

For 1 & 2

$$Q_1 = nC_p\Delta T = \frac{7}{2}nR(T_2 - T_1) \qquad \frac{7}{2}\times n\times R(T_1)$$

For 2 & 3

$$Q_2 = nC_p\Delta T = \frac{5}{2}nR(T_3 - T_2)$$

$$\frac{5}{2}\times n\times R(2T_1) = 5nRT_1$$

For 3 $\rightarrow$4

$Q_3 = 0$

For 4 $\rightarrow$1

$$Q_4 = nRT\ln\frac{V_2}{V_1}$$

$$= nRT_1\ln\frac{1}{64} = -nRT_1\ln 64$$

$$n = 1 - \frac{Q-}{Q+} = 1 - \frac{nRT_1\ln 64}{\frac{7}{2nRT_1} + 5nRT_1} \qquad = 1 - \frac{6\ln 2}{\frac{17}{2}} = 1 - \frac{12}{17}\ln 2$$

17. Total flux coming out of complete closed surface is given as

$$\phi_{hemisphere} + \phi_{cone} = \frac{q}{\varepsilon_0} \ldots\ldots\ldots\ldots (1)$$

Through the hemisphere, half of flux originated by q will come out, given as

$$\phi_{hemisphere} = \frac{q}{2\varepsilon_0}$$

Thus equation-(1), flux coming out of conical surface is given as

$$\phi_{come} = \frac{q}{2\varepsilon_0} \qquad \Rightarrow \frac{nq}{6\varepsilon_0} = \frac{q}{2\varepsilon_0} \qquad \Rightarrow n = 3$$

18. $$\Delta W = \int_{(OA)} \bar{F}.\overline{dr} + \int_{(AB)} \bar{F}.\overline{dr}$$

$$\Delta W = \int_0^4 x^3 dx + \int_0^6 12ydy$$

$$\Delta W = 152J$$

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CHEMISTRY

19. Rate of effusion Area of cross section

$$\therefore \frac{r_A}{r_B}=\frac{A_A}{A_B}=\frac{\pi(2r)^2}{r^2}=\frac{4\pi}{1}$$

20.

$$d=\frac{Z.M}{N\ a^3}$$

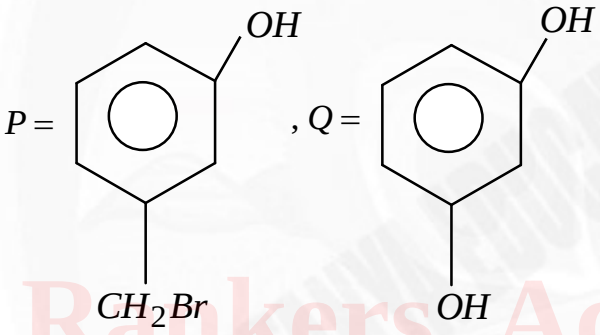
$$=\frac{4\times 6.023y}{\left(6.023\times 10^{23}\right)\times 8y\times 10^{-21}}$$

$$=\frac{1}{200}gm/cm^3=5kg/m^3$$

$$\left(\begin{array}{l}\therefore r^-+r^+=\frac{a}{2}\Rightarrow y^{1/3}=\frac{a}{2}\\ \Rightarrow \frac{a}{2}=y^{1/3}\times 10^{-7}\ cm\\ a^3=8y\times 10^{-21}\ cm^3\end{array}\right)$$

21. Tetrachlorplatinate (II) is a square plannar complex

22.



23. The correct form of (a) is
The concentration of A decreases exponentially with time according to sum of the the rate constant (k_1+k_2)

24.

$$\frac{dE}{dT}=\frac{68.4-66.3}{298-293}=\frac{2.1}{5}=0.42mv$$

$$E_{300k}=E_{298}+\frac{dE}{dT}(dT)=68.4+(0.42)(2)=69.24mv$$

$$\Delta G_{300k}=-nEE=\frac{2\times 96000\times 69.24\times 10^{-3}}{1000}=-13.29KJ$$

$$\Delta S=\frac{nFE}{dT}=2\times 96000\times 0.42\times 10^{-3}=80.64J10^{-1}$$

$$\Delta H=\Delta GT\ \Delta S=-13.29+\frac{300\times 80.64}{1000}=-13.29+24.19=10.90\,KJ$$

25.

$$BOH+HCl\Rightarrow BCl+H_2O$$

initially 5.6

When base is treated with 30ml 0.14 HCl

1.4 – 4.2 –

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For basic buffer $\therefore {}^{O}_{P}H = P^{Kb} + \log\frac{4.2}{1.4} = P^{Kb} = \log^3 = 3.3 + 0.48 = 3.78 \therefore P^H = 10.22$

At equilibrium point $(salt) = \frac{5.6}{80} = 0.07 = 7\times10^{-2}$ $\therefore p^H$ *of salt of* S_A & S_B *is*

$= \frac{1}{2}\left[14 - 3.3 - \log 7\times10^{-2}\right]$ $= \frac{1}{2}[14 - 3.3 + 2 - 0.85]$

26. SO_3 SO_2

$\mu = 0$ $\mu \neq 0$

27.

$XeF_2 + PF_5 \rightarrow [XeF]^+[PF_6]^-$

$XeF_4 + O_2F_2 \rightarrow XeF_6 + O_2$

$XeF_2 + H_2O \rightarrow Xe + HF + O_2$

$XeF_4 + 2H_2O \rightarrow \frac{2}{3}Xe + \frac{1}{3}XeO_3 + HF + \frac{1}{2}O_2$

$XeF_6 + H_2O \rightarrow XeOF_4 \rightarrow XeO_2F_2 \rightarrow XeO_3$

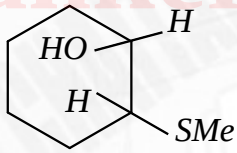
28. $C_2 - \sigma|s^2 < \sigma^*|s^2 < \sigma 2s^2 < \sigma^* 2s^2 < \pi 2p^2 x = \pi 2py^2$

Molecular orbitals are polycentric

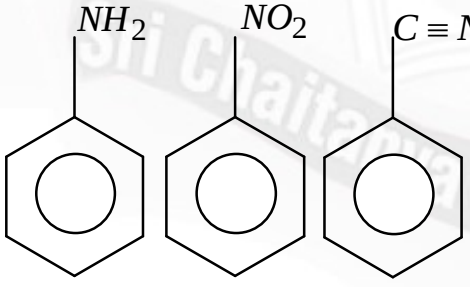
$B_2 - \sigma|s^2 < \sigma^*|s^2 < \sigma 2s^2 < \sigma^* 2s^2 < \pi 2px^{'} = \pi 2py^1$

$F_2 - \sigma|s^2 < \sigma^*|s^2 < \sigma 2s^2 < \sigma 2pz^2 < \pi 2px^2 = \pi 2py^2 < \pi 2^* px^2 = \pi^* 2py^2$

29.

$X = Y =$ 

-SMe can show NGP in the formation of X But not in formation of Y, Y is formed by S_N2

30. 

31. Cell reaction is $H_2(g) + HgCl_2(s) \rightarrow 2H^+ + 2Hg + 2Cl^-$

$E_{cell} = E^0_{cell} - \frac{0.06}{n}\log\left(H^+\right)^2 \Rightarrow 0.58 = 0.28 - \frac{0.06}{2}\log\left(H^+\right)^2$

$\therefore 0.10 = 0.06\,p^H \Rightarrow p^H = 5$

32.

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$$\lambda_1 = \frac{h}{\sqrt{2mq(100)}} = \frac{k}{10}, \lambda_2 = \frac{h}{\sqrt{2mq(81)}} = \frac{k}{9}$$

$$\lambda_3 = \frac{h}{\sqrt{2mq(49)}} = \frac{k}{7}, \frac{\lambda_3 - \lambda_2}{\lambda_1} = \left(\frac{\frac{k}{7} - \frac{k}{9}}{\frac{k}{10}} \right) = \frac{20}{63}$$

33.

$$Al(OH)_3 + NaOH \rightarrow NaAlO_2 + 2H_2O$$

$$Zn(OH)_2 + NaOH \rightarrow Na_2ZnO_2 + 2H_2O$$

$$Sn(OH)_2 + 2NaOH \rightarrow Na_2SnO_2 + 2H_2O$$

All these salts are soluble in excess of NaOH

34.

$$Cl_2 + 3F_2 \xrightarrow{573k} \underset{[A]}{2ClF_3}$$

$$\underset{[B]}{ClF_3} \xrightarrow{U} UF_6 + \underset{[C]}{ClF}$$

x = 11, y = 18, z = 6

11+18 - 6 = 15+b

b = 8

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35. 8

$O_2NCH_2COOH, HC \equiv CCH_2COOH,\ HC \equiv C - COOH, H_2C = CH - COOH,$
$PhCH_2COOH,\ Cl_3C - COOH,\ H_2C = CHCH_2COOH$

36. 8

$HC \equiv CH,\ H_2O, CH_3CH_2OH, CH_3NH_2, MeCOCH_2COMe, CH_3COOH,\ PhNH_2$
$CH_3C \equiv CH$

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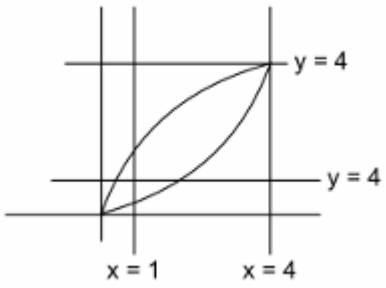
MATHEMATICS

37.

$y^2 = 4\left[\sqrt{y}\right]x \Rightarrow y \in [1,4)\left[\sqrt{y}\right] = 1$

$\Rightarrow y^2 = 4x$ *similarly* $x \in [1,4)\left[\sqrt{x}\right] = 1$

$x^2 = 4y$



Required area

$$\int_1^2 \left(2\sqrt{x} - 1\right)dx + \int_2^4 \left(2\sqrt{x} - \frac{x^2}{4}\right)dx = \frac{11}{3}$$

38.

$a + b + c = o$ .....(1)

$50 < 49a + 7b + c < 60$

$\Rightarrow 50 < 48a + 6b < 60 \Rightarrow \frac{50}{6} < 8a + b < 10$

$\Rightarrow 8a + b = 9$ .....(2)

$70 < 64a + 8b + c < 80 \Rightarrow 70 < 63a + 7b < 80 \Rightarrow 10 < 9a + b < \frac{80}{7}$

$9a + b = 11$ .....(3)

from $(1), (2)$ *and* $(3), a = 2, b = -7, c = 5$

$\Rightarrow f(x) = 2x^3 - 7x + 5 \therefore f(10) = 135 \Rightarrow k = 10$

39. Two 'a' can be filled in ${}^4C_2 . {}^4P_2 = 72$. Two 'b' can be filled in 72 ways

two 'a' and two 'b' can be filled in 72^2 ways. Now, we have to exclude the cases when

(I) Both 'a' and 'b' are in same cell i.e. 72 ways

(II) One 'a' and 'b' are in same cell i.e 72 x 16 ways

Required number of ways $= 72^2 - 72 - 72 \times 10 = 3960$

40. As equation has 4 real roots, applying Descarte's rule of sign $\Rightarrow$ equation has 4 positive real roots

$$\Rightarrow \frac{\frac{1}{x_1} + \frac{2}{x_2} + \frac{3}{x_3} + \frac{4}{x_4}}{4} \geq \sqrt[4]{\frac{24}{x_1 x_2 x_3 x_4}} = 2 \Rightarrow 2 \geq 2 \Rightarrow \frac{1}{x_1} = \frac{2}{x_2} = \frac{3}{x_3} = \frac{4}{x_4} = k$$

$$\Rightarrow \frac{1}{x_1}.\frac{2}{x_2}.\frac{3}{x_3}.\frac{4}{x_4} = k^4 \Rightarrow k = 2 \ \therefore \text{roots are } \frac{1}{2}, 1, \frac{3}{2} \text{ and } 2 \Rightarrow \frac{x4}{x_1} = 4$$

41. Curve through the intersection of S_1 *and* S_2 is given by $S_1 + \lambda S_2 = 0$

$$\Rightarrow x^2\left(\sin^2\theta + \lambda\cos^2\theta\right) + 2\left(h\tan\theta - \lambda h'\cot\theta\right)xy + \left(\cos^2\theta + \lambda\sin^2\theta\right)y^2$$

$$+\left(32 + 16\lambda\right)x + \left(16x + 32\lambda\right)y + 19\left(1 + \lambda\right) = 0$$

The above equation will represent a circle if

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$\sin^2\theta + \lambda\cos^2\theta = \cos^2\theta + \lambda\sin^2\theta$

$\Rightarrow \sin^2\theta + \lambda\sin^2\theta = \cos^2\theta + \lambda\cos^2\theta + \lambda - \lambda\cos^2\theta$

$\Rightarrow (1-\lambda)\sin^2\theta = (1-\lambda)\cos^2\theta$

$\Rightarrow (1-\lambda)\left(\sin^2\theta - \cos^2\theta\right) = 0$

$\Rightarrow \lambda = 1 \text{ or } \theta = \frac{\pi}{4}$

$h\tan\theta - \lambda h'\cot\theta = 0$

$\Rightarrow h\tan\theta = \lambda h'\cot\theta = 0$ which is satisfied if

$\lambda = 1 \text{ and } \theta = \frac{\pi}{4} \Rightarrow h = h' \text{ or } \lambda = -1, \theta = \frac{\pi}{4}$

$\Rightarrow h + h' = 0$

42. The least length = a + b

The greatest distance = a-b

43. $z_1\left(1-|z_2|\right) = iz_2\left(1+|z_1|\right)$

$\frac{z_2}{z_1} = \frac{1-|z_1|}{i\left(|z_2|+1\right)}$ is purely imaginary

$AP = \sqrt{(1-\cos\alpha)^2 + \sin^2\alpha} = 2\left|\sin\frac{\alpha}{2}\right| = 2\sin\frac{\alpha}{2} \quad \left(\because \sin\frac{\alpha}{2} > 0\right)$

44. $Similary \;\; AQ = 2\sin\frac{\beta}{2} \text{ and } AR = 2\sin\frac{\gamma}{2}$

$Now\ AP, AQ, AR \text{ are in } G.P. \Rightarrow \sin\frac{\alpha}{2}, \sin\frac{\beta}{2}, \sin\frac{\gamma}{2} \text{ are in } G.P$

$\frac{\sin\frac{\alpha}{2} + \sin\frac{\gamma}{2}}{2} \geq \sin\frac{\beta}{2} \Rightarrow \sin\frac{\alpha+\gamma}{4}\cos\frac{\alpha-\gamma}{4} \geq \sin\frac{\beta}{2}$

$also, \sin\frac{\alpha}{2}, \sin\frac{\gamma}{2} \leq \sin\frac{\beta}{2}$

45. $\frac{(x+1)^5}{\left(x^5+1\right)} = \lambda$

Use graph

46. Equation of hyperbola is $xy - y - 2x - 2 = 0$

$(x-1)(y-2) = 4$

$XY = 4$

Conjugate hyperbola XY = -4 $\frac{a^2}{2} = 4 \Rightarrow a = 2\sqrt{2}$

Focus of H' with respect to X-Y system $\equiv \left(-2\sqrt{2}, 2\sqrt{2}\right) \text{ and } \left(2\sqrt{2}, -2\sqrt{2}\right)$

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So focus x-y system $\equiv\left(1-2\sqrt{2},\ 2+2\sqrt{2}\right)$ *and* $\left(1+2\sqrt{2},2-2\sqrt{2}\right)$

Equation of transverse axis Y = X

$y-2=x-1\Rightarrow x-y+1=0$

$$1-x+x^2-x^3+.....+\left(-x^n\right)=\frac{1+(-1)^n X^{n+1}}{1+x}$$

47.

$$\Rightarrow\int_0^1\left(1-x+x^2-x^3+.....+\left(-x^n\right)\right)dx=\int_0^1\left(\frac{1}{1+x}\right)dx+(-1)^n\int_0^1\left(\frac{x^{n+1}}{1+x}\right)dx$$

$$\Rightarrow\left|X-\ln 2\right|=\int_0^1\left(\frac{x^{n+1}}{1+x}\right)dx,\ where\ \frac{1}{2(n+2)}<\int_0^1\left(\frac{x^{n+1}}{1+x}\right)dx<\frac{1}{2(n+1)}$$

$$\left(a.\,2^x+b.\,3^y+1\right)^2\geq\left(4^x+9^y+1\right)\left(a^2+b^2+1\right)$$

48. $$\Rightarrow\left(a.\ 3^y-b.\,2^x\right)^2+\left(2^x-a\right)^2+\left(3^y-b\right)^2\leq 0$$

$\Rightarrow a=2^x,\ b=3^y.$

$$\frac{1}{k}=\cos a\cos 2a\cos 3a\ .....\cos 999a$$

$$letp=\sin a\sin 2a\ .....\sin 999a$$

49. $$\therefore\frac{2^{999}p}{k}=(2\sin a\cos a)(2\sin 2a\cos 2a).....(2\cos 999a\sin 999a)$$

$$\frac{2^{999}p}{k}=2\sin 2a\sin 4a\ .....\sin 1998a$$

$$=(\sin 2a\ \sin 4a\ .....\sin 998a)(-\sin(2\pi-1000a))(-\sin(2\pi-1002\alpha))....(-\sin(2\pi-1998a))$$

$$\Rightarrow(\sin 2a\ \sin 4a\ .....\sin 1998a)(\sin 999a\ \sin 997a\ .....\sin a)$$

$$=p$$

$$\Rightarrow 2^{999}=k$$

50.

$$f'(x)=x\left(e^x-1\right)(x-1)(x-2)^2(x-3)^5$$

$f(x)\ has\ \max imum\ at\ x=2\Rightarrow k=2$

$Now\ \sin x+\cos ecx=2\ \left[\because k=2\right]$

$$\Rightarrow(\sin x-1)^2=0\Rightarrow\sin x=1$$

$$\therefore\sin^n x+\cos ec^n x=2$$

51. Line L is $\frac{x-1}{1}=\frac{y-2}{2\sqrt{2}}=\frac{z}{0}$

This line L makes 45^0 with plane $\sqrt{2}x+y-z=1$

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Required distance PQ is perpendicular distance of plane from $P=\dfrac{\left|\sqrt{2}+7-\sqrt{2}-1\right|}{\sqrt{2+1+1}}=3$

$BA=AB^2\Rightarrow A^{-1}(BA)=B^2$

$\Rightarrow\left(A^{-1}BA\right)^m=B^{2m}\Rightarrow A^{-1}B^mA=B^{2m}$

52. $\Rightarrow B^mA=AB^{2m}$

also, $BA=AB^2\Rightarrow A=B^{-1}AB^2$

$\Rightarrow A^n=B^{-1}A^nB^{2^n}$

53. Any point on $(t+2)(x+y)=1$ *is* $\left(\alpha,\dfrac{1}{t+2}-\alpha\right)$

Equation of chord of contact is $4x\alpha+16y\left(\dfrac{1}{t+2}-\alpha\right)=1$

$\Rightarrow\alpha(4x-16y)+\left(\dfrac{16}{t+2}y-1\right)=0$

The chord of contact passes through the intersection of $x-4y$ *and* $\left(\dfrac{16}{t+2}y-1\right)=0$

The point of intersection is $\left(\dfrac{1}{t+2},\dfrac{t+2}{16}\right)$

It lies inside the ellipse if $4\left(\dfrac{1}{t+2}\right)^2+16\left(\dfrac{t+2}{16}\right)^2-1<0$

$\Rightarrow\left(-\dfrac{4}{\sqrt{5}}-2\right)<t<\left(\dfrac{4}{\sqrt{5}}-2\right)\Rightarrow t=-3,-1$

54.

$f(x)=0\Rightarrow(\sin x+2+p)\left(\sin^2x-(2+p)\sin x+(p-1)^2+3\right)=0$

$\therefore$ *either* $\sin x+2+p=0$ *or* $\sin^2x-(2+p)\sin x+(p-1)^2+3=0$

$\Rightarrow p=-(2+\sin x)$ *which is false as* $p>0$

or $(\sin x-2)^2+(2-p)^2+(p-\sin x)^2=0\Rightarrow\sin x=p=2$ *but* $\sin x\neq 2$

$\therefore$ *no solutions*

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